```
from pgmpy.models import DiscreteBayesianNetwork
from pgmpy.factors.discrete import TabularCPD
from pgmpy.inference import VariableElimination

model = DiscreteBayesianNetwork([
    ('Rain', 'Traffic'),
    ('Traffic', 'Late')
])


cpd_rain = TabularCPD(
    variable='Rain',
    variable_card=2,
    values=[[0.7], [0.3]]
)

cpd_traffic = TabularCPD(
    variable='Traffic',
    variable_card=2,
    values=[
        [0.8, 0.3],
        [0.2, 0.7]
    ],
    evidence=['Rain'],
    evidence_card=[2]
)

cpd_late = TabularCPD(
    variable='Late',
    variable_card=2,
    values=[
        [0.9, 0.4],  # Not late
        [0.1, 0.6]   # Late
    ],
    evidence=['Traffic'],
    evidence_card=[2]
)

model.add_cpds(cpd_rain, cpd_traffic, cpd_late)


print("Model Valid:", model.check_model())

infer = VariableElimination(model)


result = infer.query(variables=['Late'], evidence={'Rain': 1})

print(result)
```

```
Model Valid: True
+---------+-------------+
| Late    |   phi(Late) |
+=========+=============+
| Late(0) |      0.5500 |
+---------+-------------+
| Late(1) |      0.4500 |
+---------+-------------+
```

```
#Loan Default Risk Predictor
from pgmpy.models import DiscreteBayesianNetwork
from pgmpy.factors.discrete import TabularCPD
from pgmpy.inference import VariableElimination

# Step 1: Define structure
model = DiscreteBayesianNetwork([
    ('Income_Stability', 'Default_Risk'),
    ('Credit_History', 'Default_Risk'),
    ('Employment_Type', 'Default_Risk')
])

# Step 2: Define CPDs

# Income Stability (0 = Stable, 1 = Unstable)
cpd_income = TabularCPD(
    variable='Income_Stability',
    variable_card=2,
    values=[[0.7], [0.3]]
)
```

```
# Credit History (0 = Good, 1 = Bad)
cpd_credit = TabularCPD(
    variable='Credit_History',
    variable_card=2,
    values=[[0.6], [0.4]]
)

# Employment Type (0 = Salaried, 1 = Self-employed)
cpd_employment = TabularCPD(
    variable='Employment_Type',
    variable_card=2,
    values=[[0.65], [0.35]]
)

# Default Risk (0 = Low, 1 = High)
cpd_default = TabularCPD(
    variable='Default_Risk',
    variable_card=2,
    values=[
        # Low Risk
        [0.9, 0.7, 0.6, 0.3, 0.8, 0.5, 0.4, 0.1],
        # High Risk
        [0.1, 0.3, 0.4, 0.7, 0.2, 0.5, 0.6, 0.9]
    ],
    evidence=['Income_Stability', 'Credit_History', 'Employment_Type'],
    evidence_card=[2, 2, 2]
)

# Step 3: Add CPDs
model.add_cpds(cpd_income, cpd_credit, cpd_employment, cpd_default)

# Step 4: Check model
print("Model Valid:", model.check_model())

# Step 5: Inference
infer = VariableElimination(model)

# Example 1: Full data prediction
result1 = infer.query(
    variables=['Default_Risk'],
    evidence={
        'Income_Stability': 0,
        'Credit_History': 0,
        'Employment_Type': 0
    }
)

print("\nFull Data Prediction:")
print(result1)

# Example 2: Incomplete data (missing Employment Type)
result2 = infer.query(
    variables=['Default_Risk'],
    evidence={
        'Income_Stability': 1,
        'Credit_History': 1
    }
)

print("\nIncomplete Data Prediction:")
print(result2)
```

```
Model Valid: True

Full Data Prediction:
+-----------------+---------------------+
| Default_Risk    |   phi(Default_Risk) |
+=================+=====================+
| Default_Risk(0) |              0.9000 |
+-----------------+---------------------+
| Default_Risk(1) |              0.1000 |
+-----------------+---------------------+

Incomplete Data Prediction:
+-----------------+---------------------+
| Default_Risk    |   phi(Default_Risk) |
+=================+=====================+
| Default_Risk(0) |              0.2950 |
+-----------------+---------------------+
| Default_Risk(1) |              0.7050 |
+-----------------+---------------------+
```

```
!pip install pgmpy
```

```
from pgmpy.models import DiscreteBayesianNetwork
from pgmpy.factors.discrete import TabularCPD
from pgmpy.inference import VariableElimination

# Step 1: Define structure
model = DiscreteBayesianNetwork([
    ('Fever', 'Disease'),
    ('Cough', 'Disease')
])

# Step 2: Define CPDs

# Fever (0 = No, 1 = Yes)
cpd_fever = TabularCPD(
    variable='Fever',
    variable_card=2,
    values=[[0.6], [0.4]]
)

# Cough (0 = No, 1 = Yes)
cpd_cough = TabularCPD(
    variable='Cough',
    variable_card=2,
    values=[[0.7], [0.3]]
)

# Disease (0 = No, 1 = Yes)
cpd_disease = TabularCPD(
    variable='Disease',
    variable_card=2,
    values=[
        [0.9, 0.6, 0.5, 0.2],  # No disease
        [0.1, 0.4, 0.5, 0.8]   # Disease
    ],
    evidence=['Fever', 'Cough'],
    evidence_card=[2, 2]
)

# Step 3: Add CPDs
model.add_cpds(cpd_fever, cpd_cough, cpd_disease)

# Step 4: Check model
print("Model Valid:", model.check_model())

# Step 5: Inference
infer = VariableElimination(model)

# Step 6: Query (Fever = Yes, Cough = Yes)
result = infer.query(
    variables=['Disease'],
    evidence={'Fever': 1, 'Cough': 1}
)

print("\nProbability of Disease given Fever & Cough:")
print(result)
```

```
Requirement already satisfied: pgmpy in /usr/local/lib/python3.12/dist-packages (1.0.0)
Requirement already satisfied: networkx in /usr/local/lib/python3.12/dist-packages (from pgmpy) (3.6.1)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from pgmpy) (2.0.2)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from pgmpy) (1.16.3)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (from pgmpy) (1.6.1)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from pgmpy) (2.2.2)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (from pgmpy) (2.10.0+cpu)
Requirement already satisfied: statsmodels in /usr/local/lib/python3.12/dist-packages (from pgmpy) (0.14.6)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from pgmpy) (4.67.3)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from pgmpy) (1.5.3)
Requirement already satisfied: opt-einsum in /usr/local/lib/python3.12/dist-packages (from pgmpy) (3.4.0)
Requirement already satisfied: pyro-ppl in /usr/local/lib/python3.12/dist-packages (from pgmpy) (1.9.1)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->pgmpy) (2.9.0
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->pgmpy) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->pgmpy) (2025.3)
Requirement already satisfied: pyro-api>=0.1.1 in /usr/local/lib/python3.12/dist-packages (from pyro-ppl->pgmpy) (0.1.2)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch->pgmpy) (3.25.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch->pgmpy) (4.1
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch->pgmpy) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch->pgmpy) (1.14.0)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch->pgmpy) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch->pgmpy) (2025.3.0)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn->pgmpy) (3
Requirement already satisfied: patsy>=0.5.6 in /usr/local/lib/python3.12/dist-packages (from statsmodels->pgmpy) (1.0.2)
Requirement already satisfied: packaging>=21.3 in /usr/local/lib/python3.12/dist-packages (from statsmodels->pgmpy) (26.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->pgm
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch->pgm
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch->pgmpy) (3.0.3
Model Valid: True
```

```
Probability of Disease given Fever & Cough:
+------------+----------------+
| Disease    |   phi(Disease) |
+============+================+
| Disease(0) |         0.2000 |
+------------+----------------+
| Disease(1) |         0.8000 |
+------------+----------------+
```